Elliptic functions

Gauss studied which regular polygons can be constructed by straightedge and compass alone. Rephrasing the problem yields an interesting generalisation: Given a (bounded) plane algebraic curve, can we construct n points on it with constant arc length in between? Taking the circle $(x^2+y^2=1)$ yields the original problem.
If the plane curve is defined by the equation $y=g(x)$ then the arc length is given by the integral of $\sqrt{1+g'(t)^2}\,dt$.

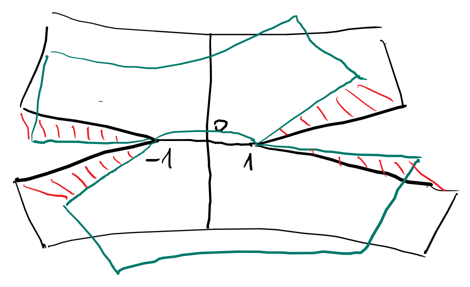
Example: The arclength of the semicircle is the integral of
$$\frac{dt}{\sqrt{1-t^2}} = \sqrt{1+g'(t)^2}\,dt,$$
where $g(t)=\sqrt{1-t^2}$.

In general, when g is a polynomial, we can consider any hyperelliptic differential, which are of the form $R(t,\sqrt{f(t)})\,dt$, for a rational function $R\in\mathbb{C}(x,y)$. If f is linear then $s=\sqrt{f(t)}$ yields a rational differential in s. In the case f is quadratic, a transformation $t\longmapsto at+b$ yields $f(t)=1-t^2$. If f is cubic or quartic, and square-free, the differential is called elliptic.

for $c^2\neq 1$

Example: The arclength of the ellipse $y^2=c^2(1-x^2)$ is the integral of
$$w=\sqrt{1+c^2\left(\sqrt{1-t^2}'\right)^2}\,dt=\sqrt{1+c^2\frac{t^2}{1-t^2}}\,dt=\sqrt{\frac{1+(c^2-1)t^2}{1-t^2}}\,dt.$$
Taking $u=1+(c^2-1)t^2$, we get $du=2(c^2-1)t\,dt$ and
$$w=\sqrt{\frac{u}{1-\frac{u-1}{c^2-1}}}\,\frac{du}{2(c^2-1)t}=\sqrt{\frac{(c^2-1)u}{c^2-u}}\sqrt{\frac{c^2-1}{u-1}}\,\frac{du}{2(c^2-1)}=-\frac{1}{2}\frac{u\,du}{\sqrt{u(u-1)(u-c^2)}}$$
which is elliptic for $R(x,y)=-\frac{1}{2}\frac{x}{y}$ and $f(x)=x(x-1)(x-c^2)$

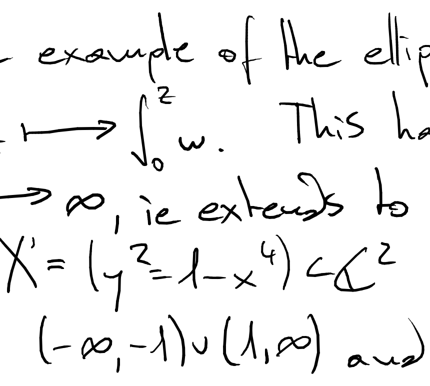
What happens if we want to consider arclength as a function? For example, consider the function
$$\phi: z\longmapsto\int_0^z w=\int_0^z\frac{dt}{\sqrt{1-t^2}}.$$
This is a function on $[0,1]$ giving the arclength of the circle. We'd like to extend ϕ to a meromorphic function. Unfortunately, for $t\neq\pm1$, there are two choices for the value of the integrand. A single-valued function ϕ can only be defined on a smaller domain, by choosing a branch.
Alternatively, define $X=(y^2=1-x^2)\subset\mathbb{C}^2$. This comes with a projection $X\to\mathbb{C}$, $(x,y)\longmapsto x$. The fibre of this projection at $x\in\mathbb{C}$ consists of the square roots of $1-x^2$. We can picture X and $X\to\mathbb{C}$ as follows: The map is generically $2:1$, with branch points ±1 (and ∞). On $Y=\mathbb{C}\setminus((-\infty,-1)\cup(1,\infty))$, there are two single-valued branches of ϕ. X is obtained by gluing two copies of this domain to each other.



It becomes clear, that X is a cylinder. The function $x\longmapsto\sqrt{1-x^2}$ on Y extends to a holomorphic function $X\longmapsto\mathbb{C}$, given by $(x,y)\longmapsto y$. Also the differential becomes $\frac{dt}{y}$, which is well-defined. Now the integral becomes a map
$$\pi_1(X,0)\times X\longrightarrow\mathbb{C},$$
or by calculating the integral along a simple loop, yielding 2π (the circumference of the circle), the integral becomes a map $X\xrightarrow{\simeq}\mathbb{C}/2\pi\mathbb{Z}$.
Note that the inverse of this map is given by $z\longmapsto(\sin z,Q(z))$. Recall that $\psi(z)=\int_0^z\frac{dt}{y}$, so that $\psi'(z)=\frac{1}{y}$. Hence, $y=\frac{1}{\psi'}=(\psi^{-1})'$, ie $Q(z)=\sin(z)'=\cos(z)$.

Let us now consider the example of the elliptic differential $w=\frac{dt}{\sqrt{1-t^4}}$. We try to define a map $\psi: z\longmapsto\int_0^z w$. This has branch points ±1 and $\pm i$. Note that ψ converges as $z\longrightarrow\infty$, ie extends to a multi-valued function $\mathbb{P}^1\longrightarrow\mathbb{C}$. We take $X'=(y^2=1-x^4)\subset\mathbb{C}^2$ and $X=\overline{X'}\subset\mathbb{P}^2$.
Taking the branch cuts $(-\infty,-1)\cup(1,\infty)$ and $(-i,i)$, we can already picture that X is a torus. The integral becomes a map $\pi_1(X,0)\times X\longrightarrow\mathbb{P}^1$. Defining the periods λ_i to be the integrals along the two simple loop, the integral becomes a map $X\xrightarrow{\simeq}\mathbb{C}/\lambda_1\mathbb{Z}+\lambda_2\mathbb{Z}$. If we write the inverse as $z\longmapsto(P(z),Q(z))$, we see again that $Q=P'$. Note that P is a meromorphic function on X, with poles at the two points corresponding to $\infty\in\mathbb{P}^1$. Note also that P satisfies the differential equation $(P')^2=1-P^4$.

The last analogy between this elliptic differential and the "circular" differential above is arc length: Define the lemniscate to be the plane curve $r^2=\cos 2\theta$.



Its arc length differential is $ds=\sqrt{dr^2+r^2d\theta}=\sqrt{1+r^2\left(\frac{\partial\theta}{\partial r}\right)^2}\,dr$. From $r^2=\cos 2\theta$, we get
$$2r=-2\sin 2\theta\,\frac{\partial\theta}{\partial r},\quad\text{ie}\quad\frac{\partial\theta}{\partial r}=-\frac{r}{\sin 2\theta},$$
hence
$$ds^2=1+\frac{r^4}{\sin^2 2\theta}=1+\frac{r^4}{1-\cos^2 2\theta}=1+\frac{r^4}{1-r^4}=\frac{1}{1-r^4}.$$
So the elliptic differential w gives the arclength of the lemniscate.

Note: Also the formula $\sin(x+y)=\sin x\cos y+\cos x\sin y$ has an analogy here:
$$P(x+y)=\frac{P(x)P'(y)+P'(x)P(y)}{1+P(x)^2P(y)^2}.$$
This is easily proved as follows: Let $g(x,y)$ denote the RHS. Let
$$h(u,v)=g\Big(\underbrace{\tfrac{1}{2}(u+v)}_{x},\underbrace{\tfrac{1}{2}(u-v)}_{y}\Big).$$
Then $\frac{\partial h}{\partial v}=\frac{\partial h}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial h}{\partial y}\frac{\partial y}{\partial v}=\frac{1}{2}\left(\frac{\partial g}{\partial x}(x,y)-\frac{\partial g}{\partial y}(x,y)\right)$. Hence $\frac{\partial g}{\partial x}=\frac{\partial g}{\partial y}$ is equivalent to $\frac{\partial h}{\partial v}=0$, ie $h(u,*)=h(u,u)$, ie $g(x,y)=g(u,0)=g(x+y,0)=P(x+y)$. So all that remains is to check that $\frac{\partial h}{\partial v}=0$, which is a calculation.